

TrES-3b

R-2738

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_150611 · created 2026-09-02T01:07:34+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457184.9169 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.056
Transit depth (Rp/Rs)^2	1.91 +/- 1.55 [%]
Orbital Inclination (inc)	80.83 +/- 2.85
Airmass coefficient 1 (a1)	0.34 +/- 0.17
Airmass coefficient 2 (a2)	0.92 +/- 1.01
Scatter in the residuals of the lightcurve fit is	49.23 %
Best Comparison Star	#2 - [486, 375]
Optimal Aperture	0.75
Optimal Annulus	3.0
Transit Duration (day)	0.034 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.138 $\pm$ 0.056 vs 0.16309 (deviation 0.45 σ)
Duration fit vs published	0.034 d vs 0.0567 d (deviation 0.81 σ)
Mid-time O–C	2.2 min
Depth SNR	2.5
Residual scatter	49.23 %

Light curve

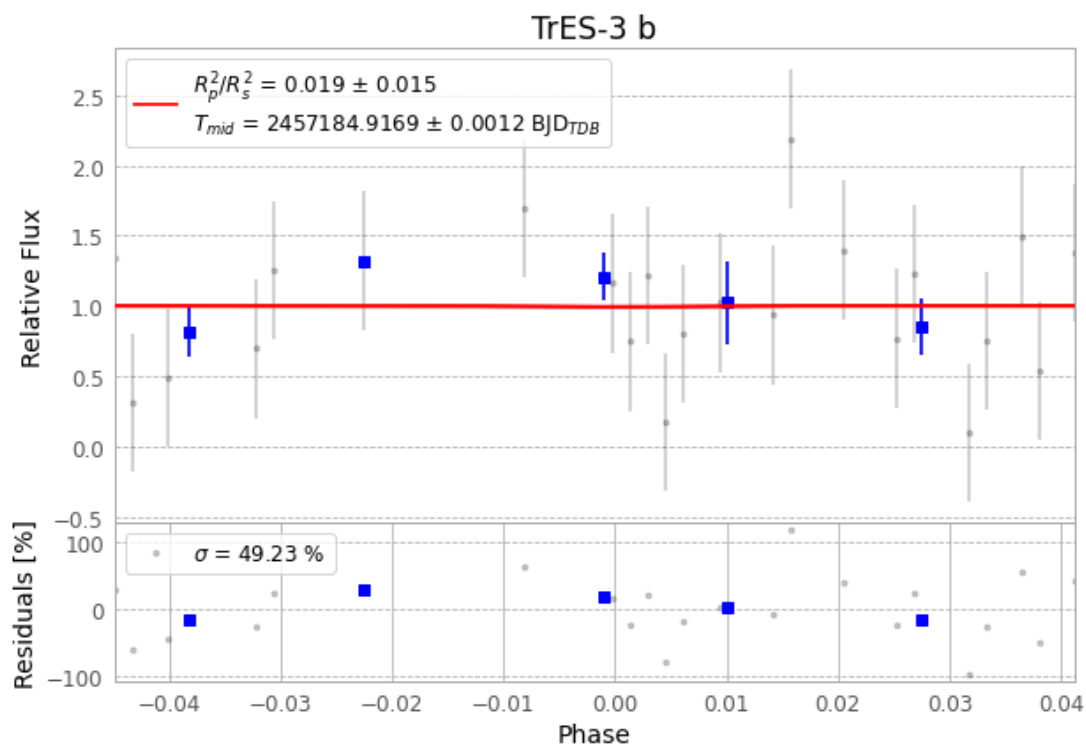


Figure 1 — FinalLightCurve_TrES-3 b_2015-06-11.png (SHA-256 3717abca660da490...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [316, 193], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d670b3480379e350...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

59 FITS frames (39 MB), TRES-3150611082112.FITS → TRES-3150611111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-150611.log (4f574f67aecb...), FinalLightCurve_TrES-3 b_2015-06-11.png (3717abca660d...), FinalLightCurve_TrES-3 b_2015-06-11.csv (42a5cb521dd3...)

Compute resources

Wall time 171.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 01:07Z.